

NITHIYANANDHAM R

Data & AI Engineer

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LinkedIn — Portfolio

Professional Summary

Proactive Data & AI Engineer with 3+ years of experience delivering real-time AI POCs, enterprise automations, and multimodal LLM solutions. Skilled in Python, OpenAI, HuggingFace, and IBM Watsonx.ai. Built Generative AI chatbots, enterprise AI platforms (ChatGPT-like), and AI workflow agents for HR and sales automation. Experienced in deploying open-source LLMs on RunPod GPU infrastructure, model fine-tuning, and building scalable API-based AI services. Developed Genesys/Twilio voice bots, multilingual STT-TTS pipelines (Sarvam, ElevenLabs), and low-latency conversational systems (less than 2s) with barge-in support. Strong in multimodal architectures using Context Memory, Context Switching, Sliding Window engines, RAG pipelines, and enterprise workflow automation. Experienced in web scraping and automated lead generation pipelines integrated with Apollo for cost-efficient sales intelligence.

Technical Skills

Programming: Python, R, SQL

AI/ML: TensorFlow, Scikit-learn, Deep Learning, HuggingFace Transformers

Generative AI / LLMs: OpenAI GPT, IBM Watsonx.ai, Granite, LLaMA, Prompt Engineering, LLM Fine-tuning

RAG & Vector Databases: FAISS, Pinecone, ChromaDB, LangChain RAG, Semantic Search

AI Infrastructure: RunPod GPU Deployment, Open-source LLM Hosting, Model Serving APIs

Voice AI: Genesys, Twilio, Sarvam TTS, ElevenLabs, Whisper STT

NLP: Text Summarization, Classification, Entity Extraction, Information Retrieval

Backend & APIs: FastAPI, REST APIs, Webhooks, Microservices Integration

Data Automation & Scraping: Web Scraping, Selenium, Automated Lead Generation, Apollo Data Workflows

Document AI: IBM Datacap, OCR Pipelines, Watson Discovery, Intelligent Document Processing

Tools & Platforms: Streamlit, Tableau, PostgreSQL, Cognos Analytics, Git, Docker

Work Experience

Data & AI Engineer — Labels Lab (US-Based, Remote)

Jan 2026 – Present

- Architected and built an **enterprise-scale AI application platform** similar to ChatGPT, enabling internal teams to access multiple AI tools through a unified interface.
- Designed and implemented **AI workflow agents** including AI HR Assistant, Sales Assistant, and Operations Copilot to automate employee support, sales workflows, and operational tasks.
- Deployed and managed **open-source LLMs on RunPod GPU infrastructure**, exposing scalable REST APIs for enterprise AI applications.
- Implemented **fine-tuning pipelines for open-source models** to improve performance on domain-specific tasks such as sales communication, customer queries, and internal knowledge retrieval.
- Built **LLM-powered tools** for automated quote generation, intelligent email drafting, design brief interpretation, and contextual task assistance for sales and design teams.
- Developed an **AI voice automation system** for inbound and outbound customer interactions, including automated sales calls, voicemail handling, and support responses.
- Created **automated lead generation pipelines** using **Apollo data and custom web scraping systems** to identify targeted prospects and generate qualified leads at **significantly reduced acquisition cost**.
- Implemented **web scraping infrastructure** to collect business intelligence, company data, and prospect information for sales automation workflows.
- Built **automated outreach systems** integrating scraped leads, Apollo datasets, and AI-generated personalized messaging for scalable sales prospecting.
- Designed **multi-agent workflow orchestration** to automate repetitive tasks across sales, marketing, and design teams, improving operational efficiency.
- Created a reusable **enterprise prompt library and AI automation framework** to standardize AI usage across departments.

- Integrated AI services with internal APIs and business systems to enable **end-to-end automation pipelines** for lead generation, follow-ups, and internal operations.
- Designed scalable **AI infrastructure and API-based microservices** to support multiple AI agents, internal tools, and enterprise workflows with high availability and cost-efficient GPU utilization.

Data & AI Engineer — SBA Info Solutions

Jan 2024 – Jan 2026

- Built Generative AI chatbots + **Voice Bots** with Genesys/Twilio handling 1,000+ queries/day.
- Delivered multilingual STT–TTS (Sarvam, ElevenLabs, Whisper) with **less than 2s response**.
- Implemented real-time **barge-in** and dynamic context switching.
- Built **RAG + VectorDB** systems using Pinecone, FAISS, ChromaDB.
- Designed multimodal memory engines (context memory, sliding window).
- Created automation agents improving operational efficiency.
- Developed Watsonx.ai predictive models .
- Automated OCR + indexing (Datacap + Discovery), reducing time by 40%.

Artificial Intelligence Intern — Imagecon India Pvt

Jan 2023 – Jan 2024

- Built ML models (image classification, NLP) with 90%+ accuracy.
- Developed Tkinter GUI tools for workflow automation.
- Created CNN models and HuggingFace NLP pipelines.

Projects

Real-Time Voice Bot with Genesys/Twilio

- Designed and deployed a production-grade voice bot integrated with Genesys Cloud and Twilio Voice, enabling automated call handling for customer support across multilingual regions.
- Built an end-to-end **less than 2 second STT → LLM → TTS** conversational loop using Whisper STT, custom OpenAI/Watsonx LLM prompts, and Sarvam + ElevenLabs TTS.
- Implemented **real-time barge-in** (interrupt detection), enabling the user to talk over bot responses without breaking the flow.
- Added conversation memory using **sliding-window memory + short-term state vectors** to maintain context across long voice calls.
- Optimized audio streaming pipeline using WebSockets to reduce buffering delay by 40%.
- Tools: Whisper STT, Sarvam TTS, ElevenLabs, OpenAI/Watsonx.ai, Twilio Voice, LangChain.

Enterprise AI Platform (ChatGPT-like Internal System)

- Built an enterprise AI platform with a ChatGPT-style interface enabling internal teams to access AI tools and workflow agents.
- Implemented AI assistants for HR and sales workflows with RAG-based knowledge retrieval.
- Deployed open-source LLMs on RunPod GPU infrastructure and exposed scalable APIs for enterprise applications.
- Tools: Python, FastAPI, LangChain, LLaMA, RunPod.

Automated Lead Generation & Sales Intelligence System

- Developed automated lead discovery using Apollo datasets and custom web scraping pipelines.
- Extracted company and decision-maker data to generate qualified B2B leads.
- Integrated AI-generated outreach messages to automate sales prospecting at low acquisition cost.
- Tools: Python, Selenium, Web Scraping, Apollo Data, FastAPI.

Open-Source LLM Hosting & Fine-Tuning Pipeline

- Deployed open-source LLMs on RunPod GPU infrastructure for scalable inference.
- Built API-based model serving for internal AI tools and automation workflows.

- Implemented fine-tuning pipelines for domain-specific enterprise tasks.
- Tools: LLaMA, HuggingFace, RunPod, Python, FastAPI.

Enterprise Document Processing Automation — IBM Datacap(For Healthcare)

- Built an end-to-end document AI workflow using IBM Datacap for OCR extraction, classification, index mapping, and automated QC.
- Automated scanning → extraction → validation → indexing pipeline for enterprise clients handling large batch documents.
- Reduced manual data entry and validation time by **40%**.
- Tools: IBM Datacap, Python, OCR Engine, IBM WatsonX.ai , IBM Watson Discovery.

Entity Extraction and Legal Document Indexing

- Developed automated indexing and legal clause extraction using Watson Discovery and transformer-based NLP models.
- Extracted 15+ legal fields including owner details, survey info, property restrictions, tenure history, etc.
- Accelerated document search by **40%** using semantic embeddings and filtered retrieval.

Legal Document AI System (Kannada–English Hybrid)

- Built a multilingual legal document pipeline capable of interpreting mixed-language land documents (Kannada + English).
- Implemented a **section-wise prompting system** using Watsonx.ai for better contextual reasoning.
- Added QC workflow with auto-flagging for low OCR confidence, missing fields, and incorrect mappings.

RAG-Based Knowledge Retrieval Platform

- Built LangChain-based RAG system using Pinecone/FAISS for hybrid search.
- Added contextual summarization, multi-document retrieval, and cited answers.
- Optimized embeddings and query ranking for long-context conversations.

AI Video Analytics & Semantic Search Platform

- Developed a video intelligence platform integrating transcription, scene-level summarization, highlight detection, and semantic search.
- Users can find video segments by meaning (not just keywords), improving discovery efficiency by **50%**.
- Tools: Whisper, HuggingFace Transformers, FastAPI, Watson Discovery.

Portfolio nithiyanandham12.github.io/Nithiyanandham.R-Portfolio/

Education

B.E. Computer Science — Mahendra Engineering College
CGPA: 7.06

2016–2020

Certifications

IBM Watson Assistant — IBM Watson Discovery — IBM Watsonx.ai — Cognos Analytics — Planning Analytics — MLOps Foundations — Instana APM — Python Fundamentals — AI using Python — HuggingFace NLP — Google Digital Marketing Fundamentals